

Bringing PCB-DESIGN And SIMULATION Together With eSim

Efficient electronic design automation (EDA) calls for performing design layout, simulation and analysis using a single tool. The open source community delivers a software for the purpose in the form of eSim. Developed by a team from IIT Bombay, eSim packs a host of features and modules that make EDA comprehensive and simple. It is built using Python and merges KiCad circuit design software and Ngspice layout simulation package, which makes it a handy tool to develop and analyse new layouts and sub-circuits, or edit existing ones.

The latest version of eSim software has a gamut of modules that work in sync to provide a compact circuit layout framework and a simulation space—all under one roof. Let's explore these different modules and how they make EDA easy.

Circuit design

Circuit design starts with schematic creation and netlist generation, followed by creating the component footprint and then finalising the circuit layout. KiCad tools like EESchema, CvPCB and PCBnew are meshed with eSim to enable circuit design.

Schematic creation with EESchema. EESchema is a popular tool for circuit schematic creation and editing. It allows you to draw the circuit and controls, which are passed down for creation of component footprints. EESchema manages the components library for eSim, enabling users to view, create or edit components. The netlist of the circuit is also developed here. It is the list of components and nodes that will be available in a circuit.



Fig. 1: eSim Startup page (Image source: esim.losse.in)

Component footprinting with CvPCB.

CvPCB completes component layout on the PCB schematic. First, it helps in assigning the correct component footprints to the schematic and netlist generated by EESchema. Then it enables the user to place components on the schematic based on the footprints. Options like footprint filtering and list viewing, and 3D view of components help to avoid errors in the process. CvPCB has the capability to assign footprints automatically with equivalence tables, thus ensuring quick and accurate footprint association.

Layout formation with PCBnew.

PCBnew works with EESchema and CvPCB to create PCB layout. Inputs are the schematic and netlist created by EESchema and PCB component layouts based on the footprint created by CvPCB. PCBnew has a plethora of features that help in creating an error-free PCB design. One such major feature is the design rules check (DRC), which points out any

erroneous arrangement in the circuit in real time. The visual interface is a rat's nest display, where components and nodes are connected by a hairline. Any movement made to the components moves the circuit along. Modifications made to the circuit schematic or components are automatically integrated by PCBnew. Effective auto-routers help in circuit production with export/import capability in SPECTRA dsn format. PCBnew also supports ultra-high-frequency circuit design.

Model framing

The eSim team has created two modules that can help users to create complex circuit models and hierarchies: Model Builder and Subcircuit Builder.

Model Builder. With the help of Model Builder, users can define the circuit model they want to build in eSim. Existing models can be edited as well. Circuit model options in Model Builder include diode, bipolar junction transistor (BJT), junction